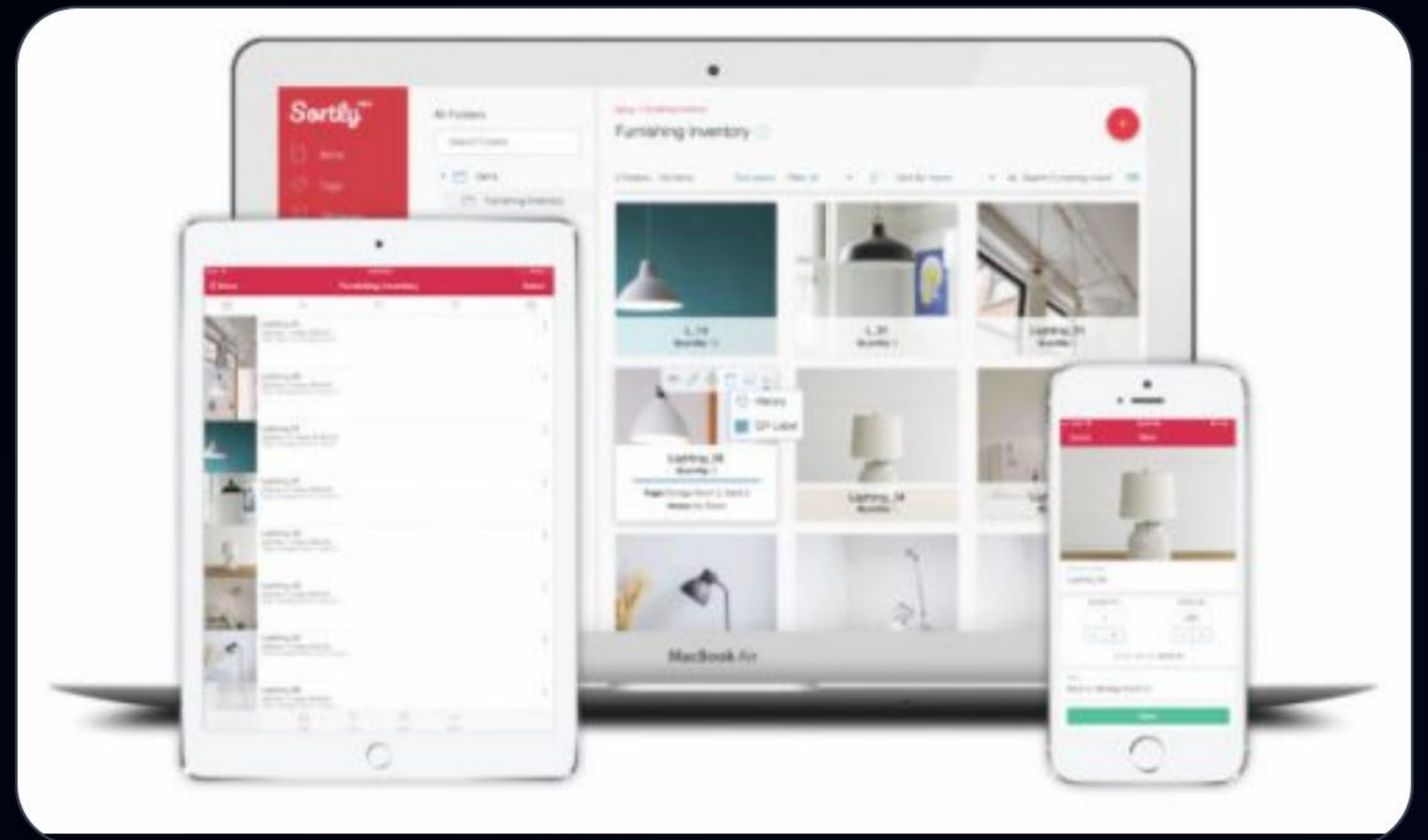


Industrial Nexus

Transitioning Factory-Floor Inventory to Local-First Architecture

| System Ecosystem

- **Admin Companion:** Web-based dashboard built with **React + Vite** for inventory oversight.
- **Worker App:** Offline-first Android tablet application built in **Kotlin**.
- **Unified Schema:** Shared data models across web and mobile via local-first synchronization.



| The "Split-Brain" Problem

Cloud Fragility

Dependency on Vercel serverless and Supabase cloud caused critical failures during factory internet outages.

Floor Bottlenecks

High-frequency operations met with socket timeouts, leading to inconsistent state between terminals.

Pure Local-First Solution



- **Local Authority:** Every terminal operates against a self-hosted Express API.
- **nexus.db**: A high-performance SQLite database serving as the source of truth.
- **Zero External Latency:** Critical floor operations complete in milliseconds without internet.

Resilient Connectivity



LAN Sync

High-speed peer-to-peer data replication over local Ethernet and WiFi.

Tailscale VPN

Secure, overlay networking for cross-site syncing and remote admin access.

No-Cloud Dependency

Synchronization continues uninterrupted even during total internet blackouts.

Rapid Terminal Pairing

Onboarding new worker terminals is simplified through dynamic QR-code pairing.

- Instant environment configuration.
- Secure handshake between Admin and Tablet.
- Zero-touch credential deployment.



| Atomic Order Promotion

Ensuring continuous workflow on the factory floor by eliminating queue stalls.

- Automatic promotion of pending orders.
- Atomic database transactions to prevent state drift.
- Fallback logic for manual override.

85%
Old Cloud-
First

12%
Industrial
Nexus

Stall Frequency (%)

| Environment-Aware Security

Security Layer	Debug / Testing	Release / Production
SSL Verification	Relaxed (Allow Self-Signed)	Strict (CA Enforced)
API Auth	Static Token	Dynamic Rotated Keys
Network Access	Open LAN	Tailscale ACL Restricted

| Reliability Gains

100%

ISP OUTAGE UPTIME

<5ms

LOCAL DB LATENCY

0

DAILY SOCKET
TIMEOUTS

Questions?

Industrial Nexus Technical Brief

Contact: tech-nexus@factory.internal

| Image Sources



Thumbnail

for

<https://camcode.com/wp-content/uploads/2021/08/sortly-e1545148076146.jpg>

Source: camcode.com



https://png.pngtree.com/png-clipart/20250424/original/pngtree-d-isolated-render-of-a-server-icon-featuring-sleek-modern-database-png-image_20786633.png

Source: [pngtree.com](https://png.pngtree.com)



<https://g4c7y7r6.delivery.rocketcdn.me/wp-content/uploads/2025/06/Hands-holding-a-tablet-showing-a-blurred-QR-code-analytics-dashboard-1024x619.webp>

Source: qrcodekit.com
